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Appendix

Use of the Conversion TABLE

$$\text{Actual gas flow rate} = \text{Output reading} \times \frac{\text{Factor of the gas}}{\text{Factor of the calibrated gas}}$$

Example 1

The controller is calibrated for nitrogen
The desired gas carbon dioxide.
The output reading is 75 sccm when carbon dioxide is flowing,
Then 75 × 0.74 = 55.50 sccm.

in order to calculate the conversion factor for a gas mixure, the following formula should be used.

$$\text{Sensor Conversion Factor Mixture} = \frac{100}{\frac{P1}{\text{Sensor Conversion Factor 1}} + \frac{P2}{\text{Sensor Conversion Factor 2}} + \frac{Pn}{\text{Sensor Conversion Factor n}}}$$

Where, P1 = percentage(%) of gas 1 (by volume)
P2 = percentage(%) of gas 2 (by volume)
Pn = percentage(%) of gas n (by volume)

Example 2

The desired gas is 20% Helium(He) and 80% Chlorine(Cl_2) by volume.
The desired full scale flow rate of the mixture is 20 slpm.
Sensor conversion factor for the mixture is

$$\text{Mixture Factor} = \frac{100}{\frac{20}{1.39} + \frac{80}{0.88}} = 0.950$$

Air equivalent flow = 20/0.950 = 21.05 slpm.

Gas Sealing Selection and Gas Factor

	Source J-836-D-508 Rev.b	Formula	GasFactor	Orifice Factor	Density (kg/m^3)	Recommended	Allowed	Not Recommended
1	1,1,2-Trichloro-1,1,2-Triflouroet (f113)	C2CL3F3	0.231	2.520	7.920	Buna	-	Viton/Buna/Kalrez
2	1,1-Difluoro-1-Chloroethane	C2H3CLF2	0.341	1.957	4.776	Buna	-	-
3	1,1-Difluoroethane	CH3CHF2	0.415	1.536	2.940	Kalrez	-	-
4	1,1-Difluoroethylene	CH2:CF2	0.458	1.512	2.860	Viton	-	-
5	1,2-Dibromotetrafluoroethane (f114B2)	C2Br2F4	0.215	2.905	10.530	Teflon	Viton/Buna/Kalrez	Epdm
6	1,2-Dichloroethane (Ethylene dichloride)	C2H4CL2	0.382	1.879	4.419	Kalrez	-	-
7	1,2-Dichlorotetrafluoroethane (f114)	C2CL2F4	0.231	2.449	7.479	Buna	Epdm	-
8	1,3-Butadiene	C4H6	0.354	1.413	2.491	Viton	Teflon-Kalrez	Buna/Epdm
9	1,1,1,2-Tetrafluoroethane (R134A)	C2H2F4	0.307	1.908	4.556	Epdm		
10	1,1,2,2,-Tetrafluoroethane (R134)	C2H2F4	0.295	1.908	4.556			
11	1,2-Propylene Oxide	C3H6O	0.348	1.440	2.594			
12	1-Butene	C4H8	0.294	1.435	2.503	Viton	Kalrez	Buna/Epdm
13	1-Pentene, 4-Methyl	C6H12	0.200	1.733	3.758			
14	2,2Dichloro–1,1,1-Trifloroethane	C2HCL2F3	0.259	2.336	6.829			
15	2,2-Dimethylpropane	C(CH3)4	0.247	1.613	3.244	Buna	-	-
16	2-Chlro-1,1,1,2-Tetrafluoroethane (R124)	C2HCIF4	0.027	2.207	6.094			
17	2-Chlorobutane	C4H9CL	0.234	1.818	4.134			
18	2-Methyl-1,3-Butadiene	C5H8	0.247	1.559	3.042			
19	3-Methyl-1-butene	C5H1O	0.252	1.584	3.127	-	-	-
20	Acetonitrile	C2H3N	0.510	1.211	1.833			
21	Acetylene(Ethyne)	C2H2	0.615	0.970	1.173	Viton	Epdm/Buna/Teflon-K	-
22	Acrolein	C3H4O	0.362	1.415	2.054			
23	Air	Air	0.998	1.018	1.293	Viton	Epdm/Buna/Teflon-K	-
24	Allene	C3H4	0.478	1.199	1.787	Buna	-	-
25	Ammonia	NH3	0.786	0.781	0.771	Epdm/Teflon	Buna/Teflon	Viton
26	Argon	Ar	1.395	1.195	1.784	Viton	Epdm/Buna/Teflon-K	-
27	Arsine	AsH3	0.754	1.661	3.478	Teflon-Kalrez	-	-
28	Benzene	C6H6	0.294	1.670	3.488			
29	Borane	H3B	0.778	0.703	0.618			
30	Boron Trichloride	BCL3	0.443	2.044	5.227	Teflon-Kalrez*	Viton	-
31	Boron Trifluoride	BF3	0.579	1.569	3.025	Teflon-Kalrez	Viton	-
32	Bromine	Br2	0.800	2.388	7.136			
33	Bromine Pentafluoride	BrF5	0.287	2.502	7.806	Teflon	Kalrez	Viton/Epdm/Buna
34	Bromine Trifluoride	BrF3	0.439	2.214	6.108	Teflon	Kalrez	Viton/Epdm/Buna
35	Bromotrifluoroethylene	C2BrF3	0.326	2.397	7.165	Viton	Buna	-
36	Bromotrifluoromethane (f13B1)	CBrF3	0.412	2.303	6.615	Buna	Epdm	Viton/Kalrez
37	Butane	C4H1O	0.257	1.467	2.593	Viton	Buna/Kalrez	Epdm
38	Carbon Dioxide	CO2	0.740	1.255	1.977	Buna	Kalrez	Viton/Epdm

※ Note : CO2 / 5bar ↑ Kalrez Set
10bar ↑ Kalrez 0 · ring Teflon seat

Gas Sealing Selection and Gas Factor

	Source J-836-D-508 Rev.b	Formula	GasFactor	Orifice Factor	Density (kg/m)	Recommended	Allowed	Not Recommended
39	Carbon Disulfide	CS2	0.638	1.650	3.393	Viton	Kalrez	Buna/Epdm
40	Carbon Monoxide	CO	0.995	1.000	1.250	Viton	Buna/Epdm/Kalrez	-
41	Carbon Tetrachloride	CCL4	0.344	2.345	6.860	Viton	Kalrez	Buna/Epdm
42	Carbon Tetrachloride (f14)	CF4	0.440	1.770	3.926	Viton	Kalrez	-
43	Carbonyl Fluoride	COF2	0.567	1.555	2.045	Viton	-	-
44	Carbonyl Sulfide	COS	0.680	1.463	2.680	Viton	-	-
45	Chlorine	CL2	0.876	1.598	3.214	Viton	Kalrez	Buna/Epdm
46	Chlorine Dioxide	CLO2	0.693	1.554	3.011	Viton	Kalrez	Buna/Epdm
47	Chlorine Trifluoride	CLF3	0.433	1.812	4.125	Kalrez	-	Viton/Buna/Epdm
48	Chlorodifluoromethane (f22)	CHCLF2	0.505	1.770	3.906	Epdm	Kalrez	Viton/Buna/Epdm
49	Chloroform (Trichloromenthane)	CHCL3	0.442	2.066	5.340	Viton	Kalrez	Buna/Epdm
50	Chloropentafluoroethane (f115)	C2CLF5	0.243	2.397	7.165	Epdm	-	Buna
51	Chlorotrifluoroethylene	C2CLF3	0.337	2.044	5.208	Teflon	-	-
52	Chlorotrifluoromethane (f13)	CCLF3	0.430	1.985	4.912	Kalrez	-	-
53	CIS-2-Butene	C4H8	0.320	1.435	2.503	Buna	-	-
54	Cyanogen	(CN)2	0.498	1.366	2.322	Kalrez	-	-
55	Cyanogen Chloride	CLCN	0.618	1.480	2.730	Kalrez	-	-
56	Cyclobutane	C4H8	0.387	1.413	2.491	Buna	-	-
57	Cyclopropane	C3H6	0.505	1.224	1.877	Buna	-	-
58	Deuterium	D2	0.995	0.379	0.177	Viton	-	-
59	Diborane	B2H6	0.448	1.000	1.235	Kalrez	-	-
60	Diboromodifluoromethane (f12B2)	CBr2F2	0.363	2.652	8.768	Viton	-	-
61	Dichlorodifluoromethane (f12)	CCL2F2	0.390	2.099	5.492	Buna	-	Viton/Teflon/Kalrez/Epdm
62	Dichloroethylene	C2H2CI2	0.397	1.860	4.329			
63	Dichlorofluoromethane (f21)	CHCL2F	0.456	1.985	4.912	Kalrez	-	Viton/Buna/Epdm
64	Dichloromethane	CH2CI2	0.522	1.741	3.793			
65	Dichlorosilane	SiH2CL2	0.442	1.897	4.506	Kalrez	-	-
66	Diethylamine	C4H1N	0.222	1.616	3.266			
67	Diedthylsilane	C4H12Si	0.183	1.775	3.940			
68	Difluoromethane (R32)	CF2H2	0.627	1.360	2.411			
69	Dimethylamine	(CH3)2NH	0.370	1.269	2.013	Kalrez	-	-
70	Dimethylether	(CH3)2O	0.392	1.281	2.055	Viton	Buna/Epdm/Kalrez	-
71	Dimethylsulfide	C2H6S	0.357	1.489	2.775			
72	Dimethylzinc	C2H6Zn	0.234	1.846	4.262			
73	Disilane	Si2H6	0.332	1.493	2.779	Teflon	-	-
74	Ethane	C2H6	0.490	1.038	1.357	Viton	Buna/Kalrez	Epdm
75	Ethanol	C2H6O	0.394	1.282	2.057			
76	Ethyl chloride	C2H5CL	0.408	1.516	2.897	Viton	Buna/Kalrez	Epdm

Gas Sealing Selection and Gas Factor

	Source J-836-D-508 Rev.b	Formula	GasFactor	Orifice Factor	Density (kg/m^3)	Recommended	Allowed	Not Recommended
77	Ethylacetylene	C4H6	0.365	1.384	2.388	Buna	-	-
78	Ethylene	C2H4	0.619	1.000	1.261	Viton	Buna/Kalrez	Epdm
79	Ethlene Oxide	C2H40	0.589	1.254	1.965	Kalrez	-	Viton/Buna/Epdm
80	Fiuorine	F2	0.924	1.163	1.695	Metal	-	-
81	Fluoroform (f23)	CHF3	0.529	1.584	3.127	Kalrez*	-	-
82	Germanium Tetrachloride	GeCL4	0.268	2.766	9.574	Kalrez*		
83	Germanium Tetrafluoride	GeF4	0.356	2.303	6.636			-
84	Germanium Tetrahydride(Germane)	GeH4	0.559	1.654	3.423			
85	Halothane (R123B1)	C2HBrCLF3	0.257	2.654	8.814			
86	Helium	He	1.386	0.378	0.178	Viton	Buna/Epdm/Kalrez	-
87	Hexafluoroacetone	C3F6O	0.219	2.434	7.414	-	-	-
88	Hexafluorobenzine	C6F6	0.632	2.577	8.309			
89	Hexafluorobutadiene	C4F6	0.213	2.405	7.236			
90	Hexafluoroethane (f116)	C2F6	0.255	2.219	6.139	Buna	-	-
91	Hexafluoropropylene (HFP)	C3F6	0.249	2.312	6.663	Buna	-	-
92	Hexamethyldisilane (HMDS)	(CH2)6Si2	0.139	2.404	7.208	Kalrez	-	-
93	Hexamethyldisiloxane	C6H18OSi2	0.110	2.408	7.251			
94	Hexane	C6H14	0.204	1.757	3.847	Viton	Buna/Kalrez	Epdm
95	Hexylamine	C6H15N	0.158	1.901	4.519			
96	Hydrogen	H2	1.008	0.269	0.090	Viton	Buna/Epdm/Kalrez	-
97	Hydrogen Bromide	HBr	0.987	1.695	3.645	Viton	Epdm/Kalrez	Buna
98	Hydrogen Chloride	HCL	0.983	1.141	1.639	Epdm	Kalrez	Buna
99	Hydrogen Cyanide	HCN	0.744	0.973	1.179	Kalrez	-	-
100	Hydrogen Fluoride	HF	0.998	0.845	0.893	Kalrez	-	Viton/Buna/Epdm
101	Hydrogen Iodide	HI	0.953	2.144	5.789	Kalrez	-	-
102	Hydrogen Selenide	H2Se	0.837	1.695	3.613	Kalrez	-	-
103	Hydrogen Sulfide	H2S	0.850	1.108	1.539	Teflon/Kalrez	Epdm	Viton/Buna
104	Iodine Pentafluoride	IF5	0.283	2.819	9.907	Teflon	-	Viton/Buna/Epdm
105	Isobutane	C4H1O	0.260	1.440	2.596	Kalrez*	-	-
106	Isobutene	C4H8	0.289	1.435	2.503	Kalrez*	-	-
107	Isopentane	C5H12	0.211	1.605	3.222	-	-	-
108	Krypton	Kr	1.382	1.729	3.708	Viton	-	-
109	Methacrolein	C4H6O	0.313	1.582	3.13			
110	Methane	CH4	0.763	0.763	0.717	Buna/Viton	Kalrez	Epdm
111	Methanol	CH4O	0.609	1.069	1.431			
112	Methyl Bromide	CH3Br	0.646	1.834	4.236	-	-	-
113	Methyl Chloride	CH3CL	0.687	1.347	2.308	Kalrez	-	Viton/Buna/Epdm
114	Methyl Fluoride	CH3F	0.761	1.102	1.518	-	-	-

Gas Sealing Selection and Gas Factor

	Source J-836-D-508 Rev.b	Formula	GasFactor	Orifice Factor	Density (kg/m^3)	Recommended	Allowed	Not Recommended
115	Methyl Mercaptan	CH4S	0.588	1.313	2.146	-	-	-
116	Methyl Silane	CH6Si	0.393	1.283	2.061			
117	Methyl Trichlorosilane(MTS)	CH3CL3Si	0.267	2.310	6.675			
118	Methyl Vinyl Ether	C3H6O	0.337	1.435	2.567	Kalrez	-	-
119	Methylacetylene	C3H4	0.473	1.196	1.782	Kalrez	-	-
120	Monoethyanolamine	C2H7NO	0.305	1.477	2.728			
121	Monoethyalamine(CH3CH2NH2)	C2H7	0.395	1.269	2.013	Kalrez	-	-
122	Monomethylamine	CH3NH2	0.565	1.067	1.420	Kalrez	-	-
123	Neon	Ne	1.398	0.847	0.902	Viton	Buna/Epdm/Kalrez	
124	Nickel Carbonyl	Ni(CO)4	0.212	2.371	7.008	-	-	-
125	Nitric Acid	HNO3	0.491	1.500	2.814			
126	Nitric Oxide	NO	0.995	1.030	1.339	Kalrez*	Viton	-
127	Nitrogen	N2	1.000	1.000	1.251	Viton	Buna/Epdm/Kalrez	-
128	Nitrogen Dioxide	NO2	0.758	1.713	2.052	Kalrez	-	
129	Nitrogen Trifluoride	NF3	0.501	1.598	3.168	Teflon	Kalrez	-
130	Nitrogen Trioxide	N2O3	0.443	1.649	3.389	-	-	-
131	Nitrosyl Chloride	NOCL	0.664	1.529	2.913	Kalrez	-	-
132	Nitrous Oxide	N2O	0.752	1.259	1.964	Buna	-	-
133	Octofluorocyclobutane	C4F8	0.169	2.672	8.933	-	-	-
134	Octafluorotetrahydrofuran	C4F8O	0.165	2.777	9.644			
135	Octofluorocyclobutane	C4F8	0.169	2.672	8.933			
136	Oxygen	O2	0.988	1.067	1.429	Viton	Epdm/Kalrez	Buna
137	Oxygen Difluoride	OF2	0.672	1.388	2.402	-	-	-
138	Ozone	O3	0.738	1.310	2.138	Viton	Epdm/Kalrez	Buna
139	Pentafluoroethane	C2HF5	0.287	2.070	5.360			
140	Pentane(n-Pentane)	C5H12	0.212	1.605	3.222	-	-	-
141	Perchlory Fluoride	CLO3F	0.448	1.905	4.571	-	-	-
142	Perfluoro-2-Butene	C4F8	0.268	2.672	8.933	-	-	-
143	Perfluorobutane	C4F10	0.738	2.918	10.610	-	-	-
144	Perfluoromethyl-Vinylether(PMVE)	PMVE	0.296	2.029	5.131	-	-	-
145	Perfluoropropane	C3F8	0.179	2.591	8.396	-	-	-
146	Phosgene	COCL2	0.504	1.881	4.418	Kalrez	-	-
147	Phosphine	PH3	0.783	1.100	1.517	Kalrez	-	-
148	Phosphorous Oxychloride	POCI3	0.327	2.340	6.847			
149	Phosphorous Pentafluoride	PF5	0.346	2.109	5.620	-	-	-
150	Phosphorous Trifluoridide	PF3	0.495	1.770	3.906	-	-	-
151	Propadiene	C3H4	0.439	1.196	1.789			
152	Propane(same as CH3CH2CH3)	C3H8	0.343	1.274	2.008	Viton	Buna/Kalrez	Epdm

Gas Sealing Selection and Gas Factor

	Source J-836-D-508 Rev.b	Formula	GasFactor	Orifice Factor	Density(kg/m^3)	Recommended	Allowed	Not Recommended
153	Propylene(Propene)*	C3H6	0.401	1.234	1.875	Viton*	Kalrez	Buna/Epdm
154	Rhenium Hexafluoride	ReF6	0.230	3.279	13.410	-	-	-
155	Silane	SiH4	0.625	1.070	1.440	Kalrez	-	-
156	Silicon Tetrachloride	SiCL4	0.310	2.462	7.579	Teflon-Kalrez	-	-
157	Silicon Tetrafluoride	SiF4	0.395	1.931	4.648	Teflon	-	-
158	Sulfur Dioxide	SO2	0.728	1.529	2.858	Epdm	Kalrez/Teflon	Buna/Viton
159	Sulfur Hexafluoride	SF6	0.270	2.348	6.516	Epdm/Teflon	Buna	Kalrez
160	Sulfur Tetrafluoride	SF4	0.353	1.957	4.776	-	-	-
161	Sulfur Trioxide	SO3	0.535	1.691	3.575		-	
162	Sulfuryl Fluoride	SO2F2	0.423	1.931	4.648	-	-	-
163	Tetrachloromethane	CCL4	0.344	2.345	6.858	-	-	-
164	Tetraethylsilane	C8H20Si	0.111	2.270	6.445			
165	Tetrafluoroethylene (TFE)	C2F4	0.361	1.905	4.526	Buna	-	Kalrez
166	Tetrafluorohydrazine	N2F4	0.367	1.926	4.624	-	-	-
167	Tetramethylsilane	C4H12Si	0.183	1.775	3.94			
168	Titanium Tetrachloride	TiCI4	0.296	2.602	8.47			
169	Toluene (C6H5)	CH3	0.234	1.814	4.115			
170	Trans-2-Butene	C4H8	0.291	1.435	2.503	-	-	-
171	Trichlorofluoromethane (f11)	CCL3F	0.374	2.244	6.281	Teflon	-	Buna/Epdm/Kalrez
172	Trichlorosilane	SiHCL3	0.329	2.201	6.038	Viton/Kalrez	-	-
173	Trifluoroethane	C2H3F3	0.333	1.732	3.753			
174	Trifluoropropene	C3H3F3	0.286	1.852	4.289			
175	Trimethyl Aluminum	C3H9AI	0.259	1.604	3.219			
176	Trimethylamine	(CH3)3N	0.316	1.467	2.639	Kalrez	-	-
177	Trimethylgallium	C3H39Ga	0.237	2.025	5.123			
178	Trimethyloxyborane (TMB)	B(OCH3)3	0.300	1.929	4.638	-	-	-
179	Trimethylsilane	C3H10Si	0.235	1.627	3.313			
180	Tungsten Hexafluoride	WF6	0.227	3.264	13.280	Teflon	-	-
181	Uranium Hexafluoride	UF6	0.220	3.548	15.700	Teflon	-	-
182	Vinyl Bromide	C2H3Br	0.524	1.985	4.772	-	-	-
183	Vinyl Chloride	C2H3CL	0.542	1.492	2.788	Viton	Kalrez	Buna/Epdm
184	Vinyl Fluoride	C2H3F	0.576	1.281	2.046	Kalrez	-	-
185	Water Vapor	H20	0.861	0.802	0.804	-	-	-
186	Xennon	Xe	1.383	2.180	5.851	Viton	Buna/Epdm/Kalrez	-
187	Z"gas mixture in volume"	mixgas	0.742	1.216	1.851			

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